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The Art and Science of Molecular Docking

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Abstract

Molecular docking has become an essential part of a structural biologist's and medicinal chemist's toolkits. Given a chemical compound and the three-dimensional structure of a molecular target—for example, a protein—docking methods fit the compound into the target, predicting the compound's bound structure and binding energy. Docking can be used to discover novel ligands for a target by screening large virtual compound libraries. Docking can also provide a useful starting point for structure-based ligand optimization or for investigating a ligand's mechanism of action. Advances in computational methods, including both physics-based and machine learning approaches, as well as in complementary experimental techniques, are making docking an even more powerful tool. We review how docking works and how it can drive drug discovery and biological research. We also describe its current limitations and ongoing efforts to overcome them.

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INTRODUCTION

The vision that one might design new drugs by using computers to assess the fit of molecules into a target binding site has inspired intensive research from generations of scientists (1, 2). This dream has, at least in part, been realized. Multiple widely used software packages have been developed to “dock” chemical compounds into a structure of a biological target, predicting how tightly each compound binds and the geometry it adopts when bound.

Today, these molecular docking methods are widely employed in biological research, drug discovery, and other fields. They are used not only to discover novel molecules that bind to proteins and nucleic acids but also for a variety of other purposes. For example, docking can help reveal the structural mechanism by which a ligand (i.e., a molecule known to bind a particular biological target) influences the target's function. Docking also frequently enables more efficient ligand optimization, a key part of drug discovery in which one begins with a ligand and searches for related compounds with more desirable properties—not only stronger binding but also improved efficacy, reduced toxicity, and fewer side effects.

In this review, we provide a high-level description of how docking works. We describe situations in which docking is particularly useful and discuss its current limitations. We provide practical guidance on how to deploy docking software and on how to use it effectively in combination with various types of experimental (wet-lab) data. We also summarize research directions that promise to substantially improve and expand the capabilities of docking methods.

Docking is most frequently used to predict interactions between protein targets and small molecules—that is, compounds with approximately 100 or fewer atoms. We therefore focus primarily on this case. We note, however, that the same methods are generally used for other

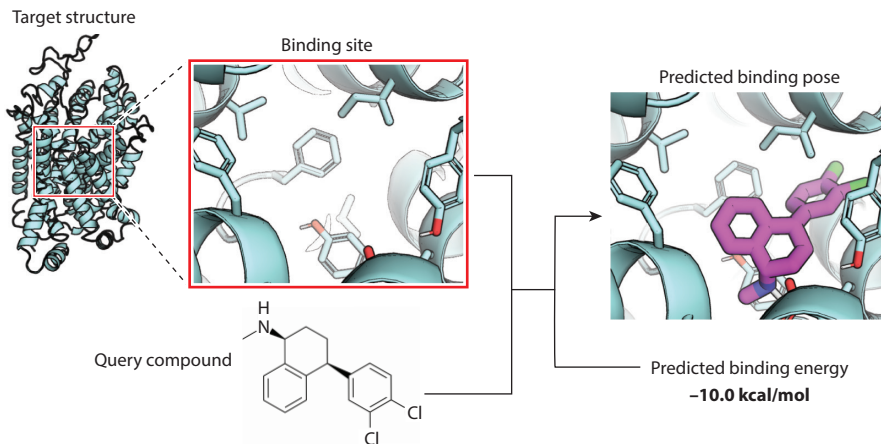


Figure 1

What docking does. Given the structure of the biological target and a compound of interest (the query molecule), docking predicts a binding pose and the associated binding energy.

macromolecular targets, particularly nucleic acids (RNA and DNA) (3, 4). These methods are also frequently applied to dock larger compounds such as short peptides (3) or macrocycles (4). Docking is also used to study protein–protein interactions, but practical considerations necessitate the use of different docking methods for that purpose (5, 6).

WHAT DOCKING DOES

Inputs and Outputs

Docking methods are designed to estimate two attributes of the interaction between a chemical compound and a biological target (**Figure 1**). First, docking predicts the binding pose of the compound—that is, the most energetically favorable geometry of the compound when bound to the target, specified by the three-dimensional (3D) coordinates of each atom in the compound relative to the target. Second, docking estimates an associated binding energy (or, equivalently, binding affinity), which quantifies the extent to which the compound binds the target as opposed to remaining in the surrounding solvent.

To this end, docking methods require the user to supply at least two inputs: a 3D structure of the target and the (2D) chemical structure of the compound. Most docking software packages also ask the user to indicate an approximate binding site, generally specified by a 3D box within which the compound is placed. This allows the user to specify the site for which they would like to discover new ligands or to supply prior knowledge about where a particular ligand is likely to bind.

Docking methods are developed such that ideally the predicted binding pose would match a corresponding structure determined experimentally by crystallography or cryogenic electron microscopy (cryo-EM), and the predicted binding energy would recapitulate the results of a binding assay, such as radioligand displacement. Docking is less accurate than these experimental methods, but it is far faster and cheaper, and it does not require each compound considered to be physically synthesized.

Importantly, the results of docking depend on the precise structure of the target that is used. The predicted binding pose and binding energy should be interpreted as estimates of the binding pose and binding energy for the particular target structure provided. If, for instance, the target

structure used for docking is in an activated state, predicted binding energies might not be relevant to an experimental assay that measures binding to an inactive state of the target.

Biophysics of Ligand Binding

Understanding docking at a quantitative level requires knowledge of the biophysics of molecular systems, particularly the relationship between potential energy, free energy, microstates, and macrostates. Readers more interested in a qualitative understanding of docking and its applications may skip to the following section.

A microstate of a molecular system specifies the precise coordinates of every atom. At body or room temperature, molecular systems are constantly fluctuating between microstates due to thermal motion of all the atoms. The potential energy, $U(x)$, of a microstate x is related to the probability, $P(x)$, of the system being in that microstate at any instant in time according to Boltzmann's equation,

$$U(x) = -k_B T \ln P(x),$$

where T is the temperature and k_B is Boltzmann's constant.

To estimate how often a compound binds to a target, one must consider all the microstates in which the compound is bound to the target and sum their probabilities to determine the total probability, $P(\text{bound})$, of the system being in a bound state at any instant in time. Such a set of microstates is called a macrostate, and its probability can be used to derive the free energy, G , of the macrostate, which is analogous to the potential energy of a microstate. For example,

$$G(\text{bound}) = -k_B T \ln P(\text{bound}).$$

It is generally most useful to consider the binding free energy, ΔG , which is simply the difference between the free energies of the bound and unbound states:

$$\Delta G = G(\text{bound}) - G(\text{unbound}).$$

(The free energy of the unbound state is computed assuming the compound is contained within a fixed volume of solvent, which is equivalent to it being at a standard concentration.)

The ligand constantly fluctuates between binding poses but typically spends most of its time in a set of very similar poses (microstates). Consequently, it is reasonable to think about a representative of this set of poses as being “the” binding pose of the ligand. Scoring functions predict the binding free energy of a ligand assuming that the given pose is representative of this set. To perform optimally, scoring functions must consider not only the potential energy of the given pose but also the number and potential energies of nearby poses (i.e., both entropy and enthalpy), as well as the set of poses the ligand adopts in the unbound state. Empirical scoring functions do this implicitly, as do machine learning–based scoring functions. Approaches based on molecular dynamics simulations explicitly consider ensembles of microstates.

Note that every chemical compound interacts to some extent with every target. The binding energy is thus a meaningful quantity for even very weakly binding compounds, typically referred to as nonbinders.

HOW DOCKING WORKS

Over a hundred software packages have been developed for docking, and many of these are widely used (7–14). In this section, we review common features of docking methods and the motivation behind various design decisions.

Docking methods generally have two components: a scoring function and a sampling procedure (**Figure 2**). The scoring function is given a hypothetical binding pose for the compound and

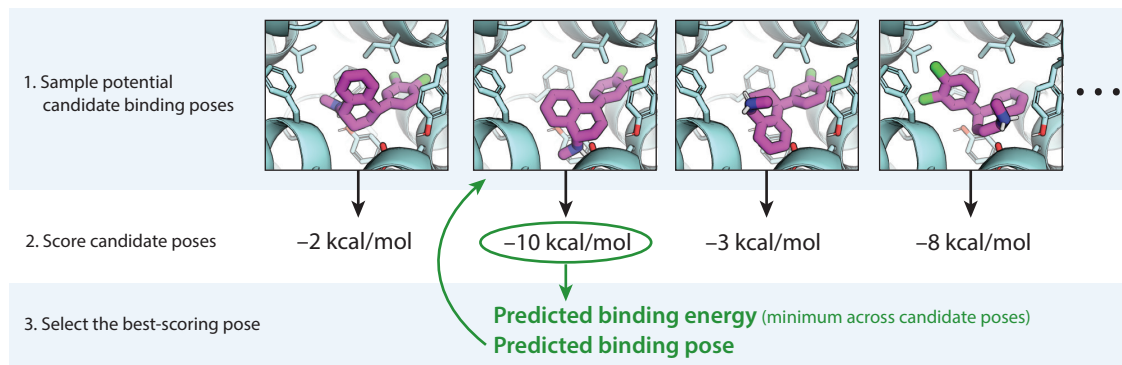


Figure 2

How docking works. Most docking methods can be decomposed into two components: a scoring function and a sampling procedure. The scoring function estimates the binding energy assuming a given binding pose, and the sampling procedure searches for the binding pose deemed most energetically favorable by the scoring function. The predicted binding pose is the one with the lowest predicted binding energy, and the estimated binding energy for the compound is the predicted binding energy for that pose.

estimates the binding energy of the compound assuming that this pose is correct. The sampling procedure searches the space of potential poses to discover the pose that is assigned the most favorable score by the scoring function. The pose with the most favorable score is the predicted pose, and the score of this pose is the predicted binding energy (**Figure 2**).

Scoring Functions

Scoring functions take as input a candidate binding pose and assess the energetic favorability of the ligand binding to the target in that pose. Most docking packages employ what are referred to as empirical scoring functions (15). In an empirical scoring function, the developer specifies terms describing various protein–ligand interactions, and the parameters in these terms are tuned such that the scoring function recapitulates experimental measurements. A variety of other types of scoring functions have been proposed (16). We focus on empirical scoring functions in this section, and we discuss scoring functions based on machine learning or molecular dynamics (MD) simulations in the section titled Ongoing Research.

Empirical scoring functions are composed of terms capturing various protein–ligand interactions known to be important for determining binding energy. The precise terms included in various scoring functions differ, but most scoring functions include terms that penalize overlapping atoms and electrostatic repulsion and reward electrostatic attraction, favorable van der Waals interactions, and hydrogen bonding. The overall score is generally computed by summing over all individual terms. For example, a hydrogen bond term would add a reward for each hydrogen bond present in the candidate binding pose. Additionally, scoring functions often include ligand-only terms to account for the entropic penalty of constraining the ligand in the binding pocket and the enthalpic penalty of the ligand adopting the shape specified by the candidate binding pose (ligand strain).

The functional form of each term is generally inspired by the underlying physics of the interaction type. However, these scoring functions are empirical in that the terms typically involve parameters that are fit such that the scoring function agrees with experimental data. Specifically, scoring functions are tuned to maximize the accuracy with which they predict binding poses, binding energies, or the best binders among a set of compounds (17, 18).

Sampling Procedures

The sampling procedure searches over various candidate binding poses to find the one evaluated to be most favorable by the scoring function. Specifically, the sampling procedure searches over the following degrees of freedom:

1. The overall position and orientation of the compound relative to the target (i.e., translations and rotations)
2. The shape of the compound itself, which can vary in many ways, particularly due to internal rotations around single bonds
3. Optionally, local rearrangement of the target structure

On a conceptual level, one can think of the sampling procedure as enumerating all possible candidate poses, evaluating the scoring function on each, and then returning the pose receiving the most favorable score, as outlined in **Figure 2**. In practice, however, most modern docking packages use scoring functions to help guide the sampling procedure. For instance, modern packages utilize search algorithms such as Markov chain Monte Carlo (9), exhaustive search using approximate scoring functions (7), or genetic algorithms (19). We refer the reader to a more detailed discussion of these algorithms (20).

In general, the sampling procedures utilized in docking do not simulate the actual binding pathway of the ligand. Doing so would be extremely slow, especially since a ligand may bind and dissociate from the receptor many times before settling into its most favorable pose. Instead, the sampling procedure moves through the space of possible binding poses in an unphysical way to discover the global minimum more quickly.

An important consideration when choosing a docking method is the trade-off between runtime and thoroughness in the search procedure. Some docking packages are designed specifically to run quickly, at the cost of less thorough sampling for individual compounds (13). This is critical for screening large compound libraries, but when one wishes to accurately predict the pose of a single ligand, runtime is much less of a concern. Some docking software packages allow the user to tune the thoroughness of sampling and thereby runtime (9).

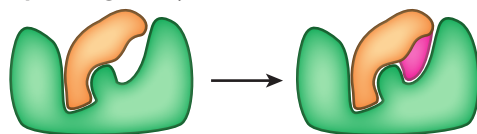
Rigid Versus Flexible Receptor Docking

Although essentially all docking software packages consider flexibility in the compounds being docked, most do not consider deformation of the target. This strategy, generally referred to as rigid receptor docking, is a major simplification. Ligand binding generally impacts the conformation (shape) of the binding pocket, with the precise binding pocket conformation depending on which ligand is bound (21, 22). Failing to take target flexibility into account can result in inaccuracies.

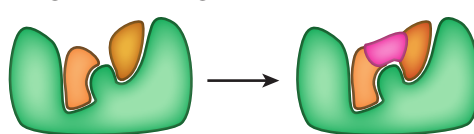
A variety of flexible receptor docking methods have been proposed (19, 23, 24) that consider target flexibility by searching over various conformations of the target binding pocket in addition to poses of the ligand. In an alternative strategy, called ensemble docking, a set of target conformations are enumerated—typically experimentally determined structures with different ligands bound—then rigid receptor docking is run on each independently, and the results are merged (25, 26).

Flexible receptor docking introduces complexity in two ways. First, considering protein rearrangements greatly increases the search space as, even just considering side chain movements in the binding pocket, the protein has many degrees of freedom. This makes flexible receptor docking orders of magnitude slower. Second, allowing the protein to rearrange requires the scoring function to evaluate the energetic favorability of various protein conformations, which is typically challenging. As a result, flexible receptor docking and ensemble docking—though useful in certain cases—do not always produce more accurate results than rigid receptor docking.

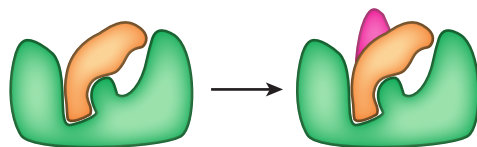
Optimizing affinity



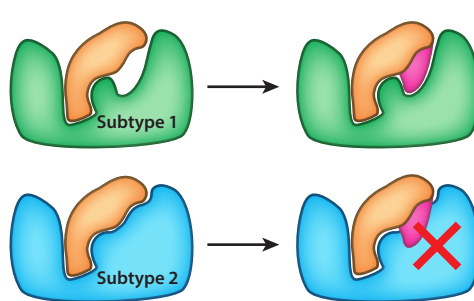
Design of bivalent ligands



Optimizing off-target properties, while maintaining affinity



Design of subtype-selective ligands



Design of labeled ligands

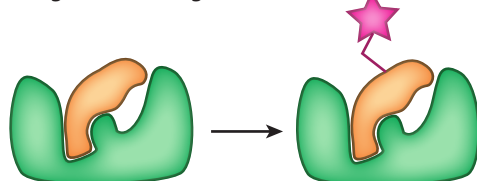


Figure 3

How docking can be used in ligand optimization. Binding poses predicted by docking can be used in much the same way that one might use experimentally determined structures to guide ligand optimization. In all cases illustrated, the predicted poses can be used either intuitively by a skilled chemist or as a starting point for more advanced computational analysis to inform ligand optimization. Pink shapes represent chemical groups added during optimization.

APPLICATIONS OF DOCKING

Ligand Optimization

Binding poses can be predicted by docking and then used in much the same way that one might use an experimentally determined ligand-bound structure to inform optimization of a ligand's binding affinity and modification of other ligand characteristics (**Figure 3**).

Predicted binding poses are often used to inform visual, intuitive analysis by medicinal chemists. Chemists can use predicted poses to assess what chemical additions, deletions, or substitutions might make the ligand more complementary to the binding site. More generally, chemists can assess which parts of the ligand might be modified to achieve a wide variety of other design goals, given the context of the binding pocket.

In addition to optimizing binding affinity, docking can inform a variety of other ligand design tasks. For example, predicted poses are used to (a) identify chemical modifications that optimize properties, such as solubility or toxicity, without degrading affinity to the target (27, 28); (b) identify locations on a ligand where chemical or fluorescent labels can be added without disrupting binding (29); (c) inform the linking of compounds to create bivalent ligands (30, 31), including compounds targeting sites on two different molecules such as PROTACs (proteolysis targeting chimeras) (32, 33); and (d) design ligands that bind selectively to one member of a set of related targets (subtype selectivity) (34–36).

In addition to intuitive analysis, automated computational methods can be used to estimate changes in binding affinity upon chemical modification of a ligand. The binding energies obtained

from docking can be used for this purpose, but they are not very accurate (37, 38). Instead poses predicted by docking are often used as starting points for other methods that can provide more accurate relative binding energy predictions, including a variety of MD simulation–based methods (see the section titled Ongoing Research).

Investigating How a Ligand Impacts Its Target's Function

Binding poses predicted by docking can illuminate the structural mechanism by which ligands affect the function of a target, and docking can thus provide fundamental insight into a target's biology. In such cases, researchers typically identify one or more ligands that elicit a functional effect of interest, dock the ligand(s) to the target, and then assess how the presence of the ligand(s) could cause the observed effects. Such approaches have been used, for example, to identify the structural basis for receptor activation. When both agonist and antagonist ligands for a receptor protein are known, the differences in their docked poses often highlight the interactions contributing to activation (39, 40). Likewise, docked poses often reveal how receptors recognize their endogenous ligands and how those ligands influence receptor function (41, 42).

Docking can also provide a starting pose for MD simulation in order to better understand the allosteric effects of a ligand (43–47). This is a powerful combination, because docking alone does not capture how the rest of the target may rearrange in response to ligand binding, and MD simulations alone are generally not able to identify binding poses. Moreover, the dynamic view of ligand–target interactions obtained through MD simulations can provide information beyond what is apparent from any static structure (48).

Discovery of Novel Ligands Using Virtual Screening

Virtual screening has emerged as a strong competitor to (experimental) high-throughput screening (HTS) for lead discovery (49, 50). In a virtual screening campaign, a library of compounds is docked, one by one, to a structure of the target, and the compounds receiving the most favorable predicted binding energies are prioritized for experimental testing (**Figure 4**). Importantly, the compound libraries used in virtual screening need only be represented on the computer and are not restricted to compounds that have already been synthesized.

This strategy has proven successful at identifying ligands that differ substantially in chemical structure from previously known ligands (50–60). Discovery of completely novel ligands can help circumvent various issues with existing ligands, such as side effects caused by off-target activity (54, 59). Moreover, many discovered ligands have unique effects on the target's function, expanding the available toolkit of drugs and chemical probes (53, 61).

Virtual screens are generally done using libraries of compounds that can be obtained cheaply without requiring custom synthesis (62). Considering sets of inexpensive compounds allows a team to purchase and test more compounds, lowering the burden for the accuracy of the predictions from docking. If a team can afford to test 100 compounds, a hit rate of 5% or even 3% is more than acceptable (see the section titled When Docking Tends to Perform Well).

Recent advances in make-on-demand libraries have greatly increased the size of the readily purchasable chemical space, making virtual screening even more important (63, 64). Make-on-demand libraries are composed of compounds that a supplier believes it can synthesize with relative ease upon request but that have largely never actually been synthesized. This strategy allows vendors to offer huge compound libraries—billions or trillions of compounds and growing—at modest prices.

Virtual screening has two main advantages over experimental HTS. First, virtual screens can consider orders of magnitude more compounds than HTS (64). Second, in virtual screening, teams

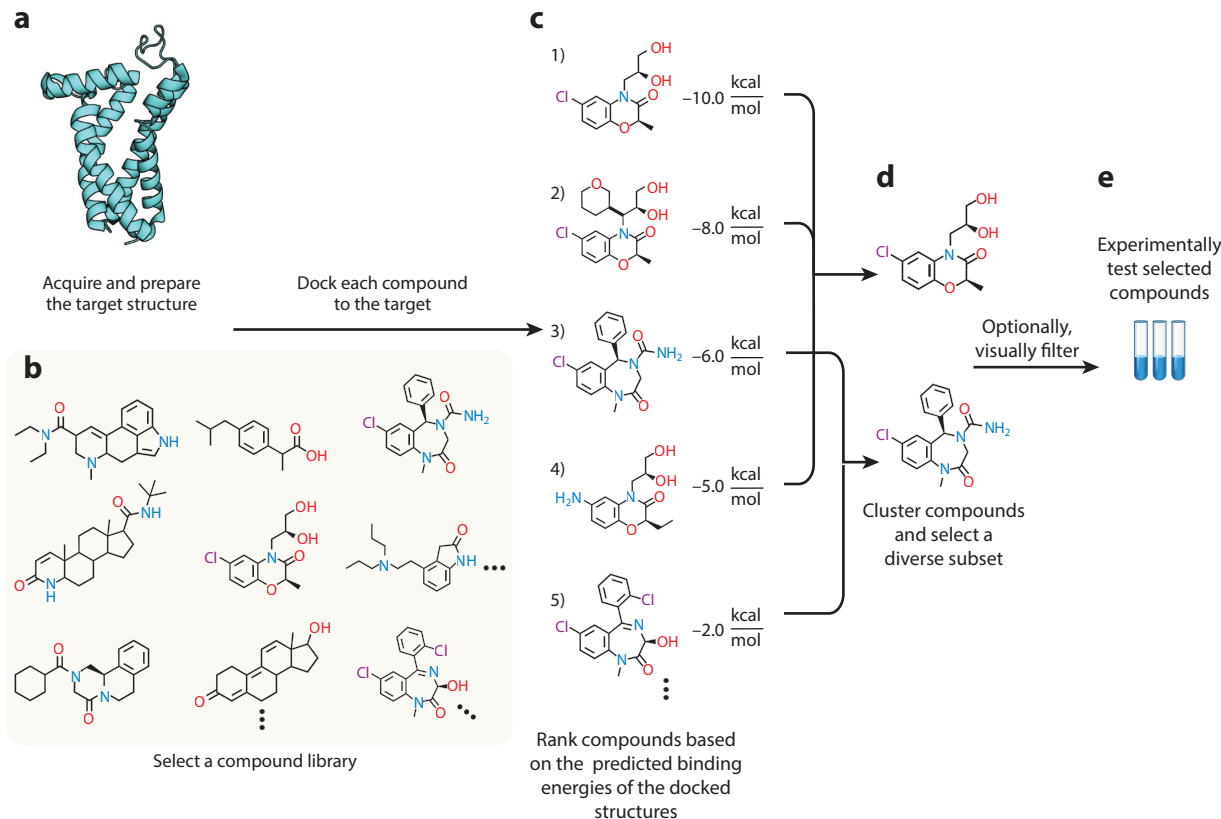


Figure 4

A typical virtual screening workflow. First, (a) the target structure and (b) the compound library to be screened are prepared. (c) Each compound is then docked to the target, and the compounds are ranked by their predicted binding energies. (d) The top-ranking compounds are clustered based on their chemical structure, and (e) a representative compound from each cluster is selected, sometimes after additional visual filtering, for experimental testing.

typically assay only tens to hundreds of compounds experimentally, so low-throughput experimental assays can be employed, circumventing the need to develop high-throughput assays suitable for HTS.

While virtual screening campaigns vary in the software and methodology employed, a typical virtual screening workflow includes the following steps:

1. Acquisition and preparation of a structure of the biological target: The structure is prepared by, for instance, adding hydrogens and selecting tautomeric states. A target binding site is also selected.
2. Selection of a compound library: One selects a compound library to dock, generally by obtaining a list of compounds available from a vendor. In some cases, knowledge of chemical features likely to be important for binding (e.g., presence of a negatively charged group to complement a positively charged group in the binding pocket) is used to inform library selection. The compounds may also be filtered to remove any compounds that are chemically similar to previously known binders.
3. Docking and ranking: Each candidate compound is docked to the target structure, and the compounds are ranked by their predicted binding energies.

4. Clustering: The top-ranking compounds are clustered based on their chemical structure, and a representative member of each cluster, generally the one with the most favorable predicted binding energy, is considered for experimental testing. This is done to avoid experimental testing of many close analogs, because close analogs are likely to either all succeed (bind) or all fail.
5. Visual filtering (optional): Some features of protein–ligand interactions are not well captured by most docking software, and it can often help to perform additional intuitive filtering (81). Generally speaking, there are three categories of things to look out for: (a) compounds in unlikely tautomeric states, such as the presence of a negatively charged hydroxyl group not ligated to a metal; (b) compounds in unlikely poses—for example, containing chemical groups that could act as hydrogen bond donors but are not doing so; and (c) compounds that form very different interaction patterns from known binders, such as the lack of a salt bridge known to be critical for binding.
6. Experimental testing: Selected compounds are procured and tested experimentally for binding to the target. Most virtual screening studies consider tens to hundreds of compounds in their primary experimental testing. It is essential to choose an experimental assay that can detect weak binders, as the initial hits are often weak binders.
7. Optimization of hits: The initial hits from docking are typically optimized to improve their affinity, either by computational evaluation of related compounds or by traditional medicinal chemistry approaches.

We note that if some ligands for a target are already known, ligand-based methods can be used as an alternative to docking for virtual screening. Roughly speaking, these methods rank compounds by their chemical similarity to known ligands. Docking-based virtual screening generally identifies more chemically diverse binders than ligand-based methods and is thus particularly valuable when one wishes to identify ligands that differ substantially from previously known ligands (65).

Other Applications

Docking can be used to search for target proteins through which a compound might act (66, 67). This task is referred to as either target fishing or reverse docking. Target fishing has been used to identify targets of natural products (68), metabolites (69), and drugs (70). Conversely, docking-based virtual screening can be used to identify endogenous or dietary compounds that might bind a target of interest (71, 72).

Docking can also be used to aid in placing ligands when solving crystal and especially cryo-EM structures—that is, going from the raw experimental data to the 3D atomic coordinates. In some cases, the experimental data on their own cannot unambiguously determine the pose of the ligand, so a pose is selected that is both consistent with the raw experimental data and scored favorably by docking (73).

WHEN DOCKING TENDS TO PERFORM WELL

Broadly speaking, docking performs best given a high-resolution target structure with a ligand bound in the desired binding site. In such cases, the binding site is well shaped to accommodate a ligand, making it easier for docking to position and score a compound of interest.

Docking to binding sites that do not contain a ligand in the target structure is more challenging. In the absence of a ligand, the binding pocket often collapses, not leaving sufficient space for most

ligands to fit (21). Moreover, functional groups, like hydrogen bond donors, may not be oriented to form favorable ligand interactions.

When Is Binding Pose Prediction Reliable?

Especially confident binding-pose predictions can be made when an experimentally determined binding pose of a ligand similar to the compound of interest is available. Ligands that share a common scaffold generally position their shared scaffold similarly in the binding pocket. An experimentally determined binding pose of a ligand sharing a scaffold with the compound being docked can thus be used to inform placement of the shared scaffold, and only the positions of the unique parts of the compound need to be predicted *de novo*. Most modern docking packages support the restraint of a common substructure during docking. Although generally quite effective, this strategy does not guarantee success: In some cases, the shared scaffolds of similar ligands adopt different positions, especially if one or both ligands are weak binders (74).

Outside of this setting, docking on its own cannot be counted on to consistently select an accurate pose (75, 76). However, most docking packages can output a ranked list of the best-scoring poses, and this list usually includes an accurate pose. One can thus consider agreement with various forms of experimental data to select an accurate binding pose from this list (see the section titled Incorporating Experimental Data).

Predicting binding poses using computational models of protein structures can be challenging, even with the advent of high-accuracy structure prediction methods such as AlphaFold (77, 78). Unlike unliganded structures, these models do not generally have collapsed binding pockets, but subtle inaccuracies in the modeled binding pocket often prevent accurate binding-pose prediction.

In some cases, flexible receptor docking is required to even sample a correct binding pose. However, it is hard to make any definitive statement about when rigid or flexible receptor docking might give more accurate results. As a rule of thumb, one might favor flexible receptor docking in cases where the binding pocket is expected to deform substantially upon ligand binding (19). Such cases include docking to unliganded binding sites and cases where the ligand of interest binds preferentially to a different target conformation than the ligand present in the structure. One might also use flexible receptor docking if none of the ligand poses proposed by rigid receptor docking is consistent with experimental data on that ligand.

For performance comparisons between specific docking packages, we refer the reader to recent benchmarking studies (79, 80).

When Does Virtual Screening Tend to be Effective?

In most virtual screening campaigns, a small number of the best-scoring compounds are tested experimentally, and success is measured by whether a few of these compounds in fact bind. To achieve success, a docking method does not need to assign favorable scores to all binders in the library, but it must rank at least a few binders above nearly all the nonbinders.

Rigid receptor docking is almost always used in virtual screening, not only because it is faster than flexible receptor docking but also because it has a lower false-positive rate—that is, it is less likely to assign favorable scores to nonbinders. Flexible receptor docking generally produces substantially more false positives because of uncertainty in the relative energetic favorabilities of the different target conformations considered.

Successful virtual screening requires that one can identify and prepare a structure of the target binding site that is well shaped to accommodate a ligand. This is best achieved by using a high-resolution structure with a ligand bound in the desired binding site. In this case, virtual screening is very likely to produce one or more novel binders when a hundred or so compounds are selected

for experimental testing. Screening against unliganded structures and computationally predicted models is more challenging, but several studies have reported success (81–87).

Most successful virtual screening campaigns have targeted deep, enclosed binding pockets (53, 61). The much smaller number of studies targeting flat, solvent-exposed pockets, as are often present at protein–protein interfaces, suggests this is more challenging (88).

The success of a virtual screening campaign is often reported as the hit rate, the fraction of experimentally tested compounds that in fact bind the target. One can express the hit rate as a product of two components: (a) the base rate at which arbitrarily chosen compounds from the library bind to the target and (b) the extent to which the docking procedure can accurately distinguish true binders from nonbinders and thereby outperform the base rate. This second component is commonly called the enrichment factor (the ratio of hit rate to base rate) and is a commonly used performance metric for virtual screening methods. The enrichment factor can be increased by using a high-quality structure, employing accurate docking methods, and leveraging any available experimental data (see the section titled Incorporating Experimental Data). Virtual screening campaigns generally also benefit from skilled chemists visually inspecting the results and filtering the list of compounds to be tested experimentally (49, 89). However, virtual screening still generally has a low (or zero) hit rate when attempting to target unfavorable binding sites. For deep, enclosed binding pockets, hit rates of a few percent are typically achievable given a high-quality structure—far better than would be achieved by testing arbitrarily chosen compounds.

Docking is typically unable to rule out the possibility that a compound binds a target, as the binding pocket might rearrange in a manner that is difficult to anticipate. This is acceptable given the typical goals of virtual screening but makes it challenging to directly apply docking to identify selective ligands that do not bind to other proteins related to the target (90). Likewise, using docking to identify ligands that bind only to one conformational state of a target has proven challenging.

We refer the reader to a more extensive practical guide on virtual screening (49).

INCORPORATING EXPERIMENTAL DATA

In most cases where one wishes to predict ligand-binding poses or perform virtual screening, some experimental data about the target are available beyond the structure used for docking. Various forms of experimental data can be used to improve the accuracy of docking results. This is most frequently done in an intuitive fashion by researchers performing docking, but some automated approaches for leveraging experimental data in particular circumstances are available. Continued development of principled, automated methods for leveraging experimental data is an important area for future work.

If experimentally determined structures of the target of interest in complex with other ligands are available, these structures can be leveraged in both pose prediction and virtual screening. The underlying theory is that most binding pockets contain hot spots that form particularly favorable interactions with certain ligand chemical groups (91), and different ligands binding to the same target frequently interact with these same hot spots. When examining the candidate binding poses ranked highly by docking software, experienced users thus typically select binding poses that form similar interactions to those seen in experimentally determined poses of other ligands. Similarly, when selecting which compounds from a virtual screen to assay experimentally, one might prioritize compounds that could form similar interactions to those seen in experimentally determined poses of other ligands. These strategies can be applied quite generally, as many different chemical groups can form similar interactions with the target (92).

Information on other compounds that bind to the target can be useful even in cases where no structures with these compounds bound are available. For example, one can tune the parameters in a virtual screen—including the docking method employed, the protonation states of target residues, and even which target structure is used—to maximize the ability of the docking software to distinguish between known binders and nonbinders (49, 81, 85). When predicting the binding poses of a congeneric series of ligands sharing a common scaffold, one might require that the common scaffold is placed in the same position for all the compounds, as illustrated by the RosettaLigandEnsemble method (93).

We recently introduced an automated method that significantly improves the accuracy of binding pose prediction given a few other ligands known to bind the target, even when those compounds do not share any common scaffold with the ligand of interest (94) (**Figure 5a**). ComBind predicts the binding poses of all known ligands simultaneously, without assuming that ligands adopt similar poses or form similar interactions with the binding pocket. Instead, ComBind uses a scoring function that considers both the similarity between the interactions formed in the predicted pose of each ligand and the standard docking scores of each individual pose. To quantify the similarity of interactions and how this should be weighted relative to the standard docking scores, we applied a machine learning strategy to a large set of data on diverse targets. This strategy robustly increases the accuracy of binding-pose prediction, even at targets very different from those used in training and even when only one or two other ligands are known (**Figure 5b**).

A related method, ComBindVS, achieves better virtual screening performance than standard docking (or ligand-based virtual screening methods) given a few ligands known to bind the target. ComBindVS does not require any experimentally determined binding poses. It significantly increases virtual screening performance even when only one or two ligands are known and even when these known ligands are very different from any compound in the virtual library (**Figure 5c**).

Mutagenesis experiments are also used to inform binding pose prediction, as mutations to protein amino acid residues that directly interact with a bound ligand typically have the largest effect on the ligand's binding affinity (95). Interpretation of mutagenesis results is complicated by the fact that some mutations have nonlocal effects on ligand binding, as well as on protein function. A strategy called mutant cycle analysis mitigates this issue by considering the effects of a mutation on a pair of ligands that differ only in one position (i.e., one small part of each ligand) (96, 97). If mutating a particular residue has substantially different effects on the binding affinities of these two ligands, the residue is most likely in contact with one or both ligands at the position where they differ.

ONGOING RESEARCH

The performance of docking methods is well past the bar required to be practically useful, but there remains significant room for improvement. A variety of ongoing work promises to increase the performance and applicability of docking.

Molecular Dynamics Simulation Approaches

A major challenge with classic docking approaches is that each score is calculated based on a single ligand pose and target conformation (see the section titled Biophysics of Ligand Binding). Even flexible receptor docking considers only one target conformation at a time during scoring. In reality, molecular systems are constantly in motion, and binding poses and energies depend on the entire ensemble of structures the system can adopt.

MD simulations, which simulate the motions of all atoms in the molecules of interest (48), can improve the accuracy of docking, at the cost of requiring significantly more computational

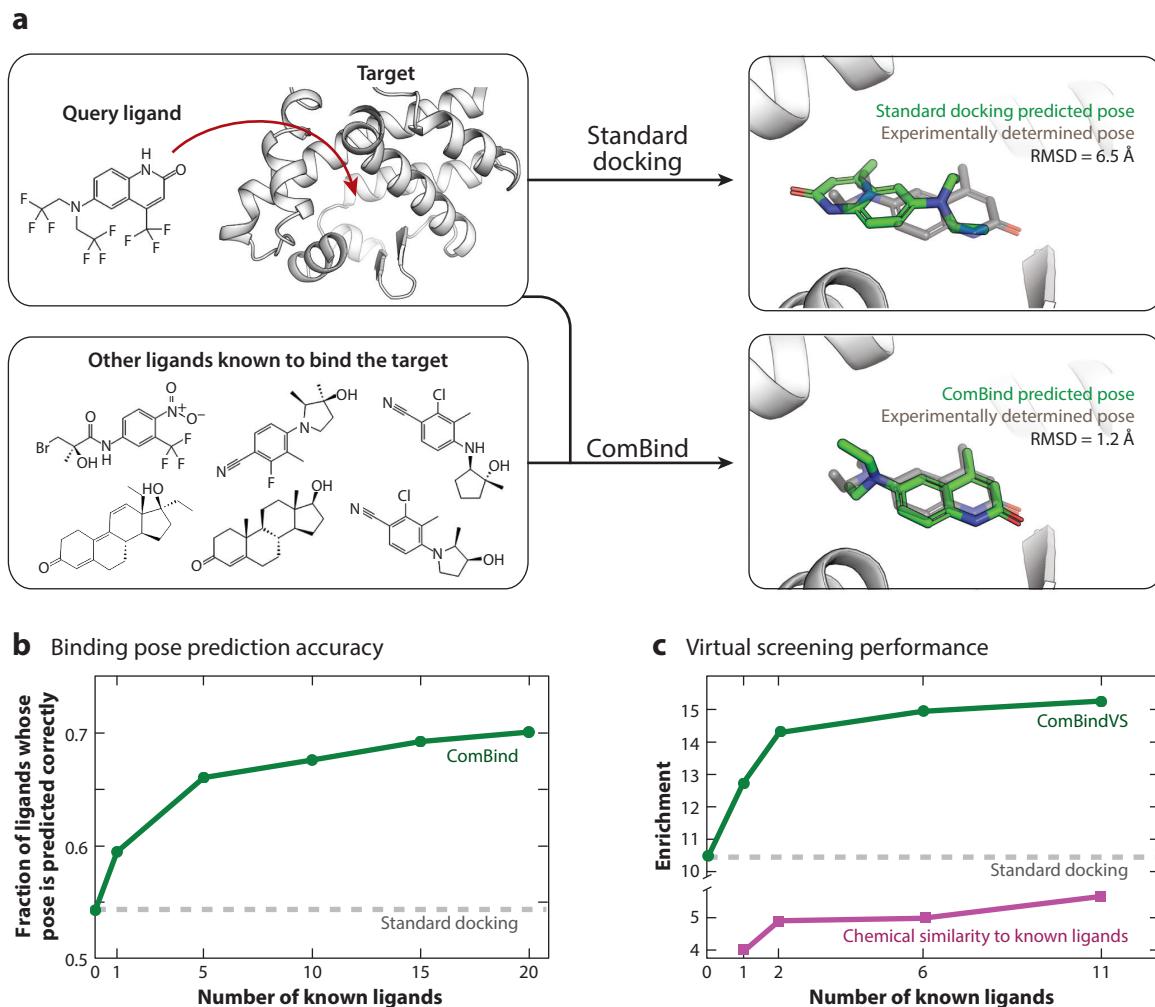


Figure 5

Incorporating knowledge of other ligands that bind the target. (a) The ComBind method for pose prediction differs from standard docking in that, when making a prediction for a query ligand, it also considers other ligands that are known to bind the target (whose binding poses are not known). (b) ComBind substantially improves binding pose prediction accuracy, even with very few additional known ligands. This plot shows the average performance across 30 diverse target proteins representing all families of common drug targets. (c) ComBindVS utilizes the same information—a few previously known ligands—to improve the accuracy of virtual screening. ComBind is first used to predict poses for the known ligands, and then the ComBind score (including the known ligands) is used to rank compounds in the candidate library. This plot shows the average performance across the DUD-E benchmark set, restricting to candidate compounds that are very different from the known ligands. ComBindVS substantially outperforms both standard docking and a ligand-based screening method that considers the chemical similarity of candidate compounds to known ligands. Abbreviations: DUD-E, Directory of Useful Decoys: Enhanced; RMSD, root-mean-square deviation. Figure adapted from Reference 94.

resources. MD simulations are becoming more accessible, thanks to the widespread availability of graphics processing units, and more accurate, thanks to improvements in the quality of the molecular force fields these simulations employ (98, 99).

Perhaps the most intuitive way to use MD simulations to predict a binding pose and energy would be simply to run a long simulation of the compound and target and observe where the

compound binds and how often it is bound. Despite some notable successes (100–103), traditional equilibrium MD simulations are typically unable to reach the timescales necessary to directly observe the ligand repeatedly binding and unbinding from the target. Several specialized simulation approaches have been developed to overcome this obstacle.

The most widely used of these specialized MD simulation approaches are “alchemical” methods that estimate the difference in binding free energy between a pair of similar ligands (104). These methods require prior knowledge of binding poses, but when applied with care, they can predict the relative binding energies across a family of structurally similar ligands much more accurately than traditional docking scoring functions. These approaches have a proven record of successfully guiding lead optimization (105). Related methods have been developed to estimate absolute binding free energies of individual ligands given their binding poses (106, 107). These methods may be used to rank chemically diverse ligands in virtual screening, but they are substantially less accurate and more computationally intensive than those used to compute relative binding energies of similar ligands. MD simulation–based methods suitable for binding pose prediction have been proposed (108, 109). These methods use customized simulation procedures that are much less computationally expensive than alchemical methods to assess the stability of candidate binding poses, making it feasible to evaluate a useful number of candidate poses.

Various strategies have also been proposed to use simulations to gain insights into favorable binding sites on a target that can then be leveraged in classic docking scoring functions. Simulations can be used to identify the favorability of displacing waters from specific sites in a binding pocket, which can be added as an additional term in scoring functions (110–112). A family of approaches, generally referred to as mixed-solvent MD simulations, utilize simulations in the presence of various organic solvents to detect favorable binding sites, including cryptic sites that are not open in available structures of the target (113, 114).

Machine Learning Approaches

A great deal of current research focuses on applications of machine learning to improve the accuracy and applicability of docking. This work holds promise in several areas.

First, machine learning scoring functions (MLSFs) aim to improve upon the accuracy of the empirical scoring functions used in traditional docking (75, 115–120). Whereas empirical scoring functions employ predefined terms inspired by the underlying physics and typically contain at most a few dozen parameters that are fit to recapitulate experimental data, the recent wave of MLSFs employ more general models (e.g., neural networks) with far more parameters that are fit to data. MLSFs generally take as input a simple representation of a candidate binding pose, such as the 3D coordinates and element types of the atoms, and then learn the scoring function directly from data. While it may seem unwise to disregard knowledge of the underlying physics, there are many inherent complexities of ligand binding that might be hard to capture with traditional scoring function terms, such as the need for the model to account for entire families of nearby poses (entropy; see the section titled Biophysics of Ligand Binding) and nonadditivity of interaction energies (121). MLSFs, for example, may provide a way to take receptor flexibility into account implicitly (122, 123). The development of MLSFs has been facilitated by efforts to assemble appropriate training and benchmarking datasets (117, 124, 125).

Second, approaches have been developed to accelerate virtual screening by approximating the docking process using machine learning models (126–128). This work is timely, as the largest make-on-demand libraries are becoming too large to screen directly. In this active-learning paradigm, a small fraction of the virtual library is docked, and the results are used to train a machine learning model that can then be quickly applied to the remainder of the library. This strategy

can also be used to approximate the results of computationally expensive alchemical methods to accelerate lead optimization (129, 130).

Third, various generative modeling approaches directly output compounds compatible with a target structure (131). Chemical-space docking methods search through combinatorial make-on-demand libraries by first docking the individual components and then expanding those that score best (132, 133). Genetic algorithms have been proposed to search for compounds with favorable docking scores (134). Deep learning approaches can also be used to generate ligands given the structure of a target binding pocket (135).

Finally, generative models for binding-pose prediction directly output binding poses, challenging the decomposition of the docking problem into sampling and scoring modules (136–138). Given the nascency of this research direction, it is not yet clear how practically useful these approaches will prove (139, 140), but initial reports show promise, especially for cases where the approximate target binding site is not known (137). Challenging the sampling and scoring paradigm in a different way, other machine learning approaches predict a binding energy without sampling poses at all (141, 142).

CONCLUSION

While its accuracy is far from perfect, docking has led to important discoveries in a variety of settings. Ongoing methodological advances promise to improve the accuracy of docking, and complementary innovations in experimental methods will make docking even more useful. Together, these advances promise to expand the capabilities of structure-based drug discovery, perhaps one day enabling the routine design of ligands with strong binding affinities, desired effects on the target's function, and favorable safety profiles.

DISCLOSURE STATEMENT

R.O.D. serves on the Scientific Advisory Boards of Septerna, Atomwise, and Atomic AI. Stanford University has filed patent applications related to methods described in References 94 and 135.

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